

Problem 3:

Known:

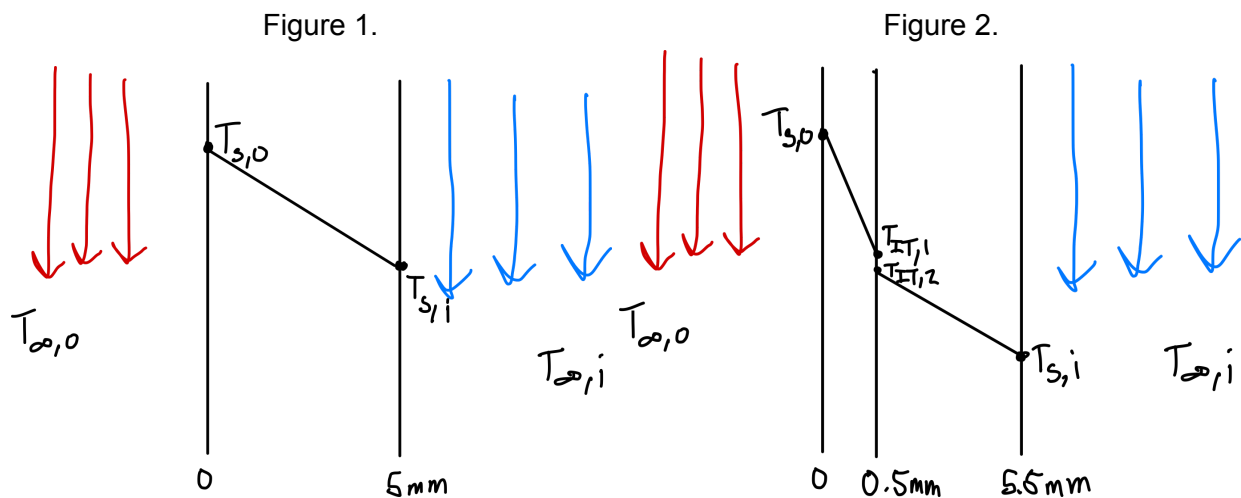
A blade of a turbine is made with a thermal barrier coating (TBC). The alloy of the blade has thermal conductivity k_b and the coating has thermal conductivity k_{TBC} . To bind the TBC to the blade a metallic bonding agent of negligible thickness is needed. This creates an interfacial thermal resistance R'' . External gas temperature is $T_{\infty,o}$ and internal gas temperature is $T_{\infty,i}$. Each gas has a heat transfer coefficient h_o and h_i for external and internal respectively. The alloy thickness is L_b . The max temperature of the alloy is T_{max} .

Find:

(a) There is no TBC. Draw the thermal resistance network and label all thermal resistances. What is the external temperature of the blade? Does this work with T_{max} ?

(b) TBC is applied. L_{TBC} is the thickness of the TBC. Draw the thermal resistance network and label all thermal resistances. What is the temperature of the blade? Does this work with T_{max} ?

Schematic:



Properties:

$$k_b = 25 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$$

$$k_{\text{TBC}} = 1 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$$

$$T_{\infty, O} = 1616 \text{ K}$$

$$T_{\infty, i} = 400 \text{ K}$$

$$R'' = 10^{-4} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$$

$$h_o = 1000 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$$

$$h_i = 500 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$$

$$L_b = 5 \text{ mm}$$

$$T_{\text{max}} = 1200 \text{ K}$$

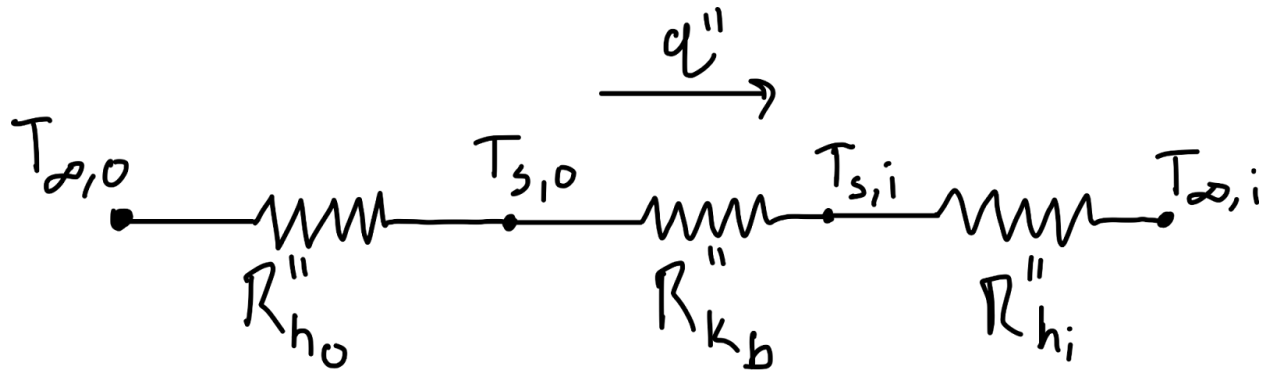
$$(b) L_{\text{TBC}} = 0.5 \text{ mm}$$

Assumptions:

Steady state, no radiation, blade can be treated as a 1D wall.

Analysis:

(a) Using the information in figure 1 the thermal resistance network below can be set up.



The thermal resistances can be calculated by:

$$R''_{h_o} = \frac{1}{h_o} = 0.001 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R''_{k_b} = \frac{L_b}{k_b} = 2 \cdot 10^{-4} \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R''_{h_i} = \frac{1}{h_i} = 0.002 \frac{\text{m}^2 \text{K}}{\text{W}}$$

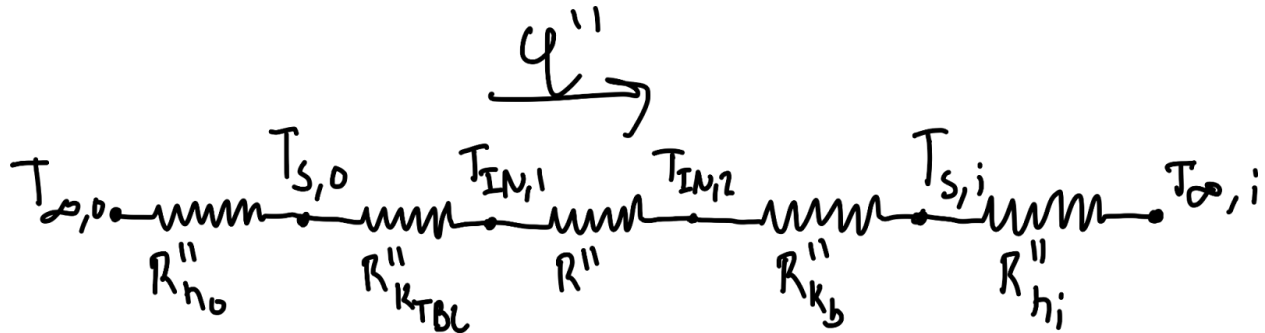
R''_{tot} is equal to the summation of all of these thermal resistances as they are all in series. This

means that $R''_{\text{tot}} = 0.0032 \frac{\text{m}^2 \text{K}}{\text{W}}$. Using the fact that $(T_{\infty, O} - T_{\infty, i}) = q'' \cdot R''_{\text{tot}}$ q'' can be

calculated to be $380 \text{ kW} \cdot \text{m}^{-2}$. Using this the equation $(T_{\infty, O} - T_{s, O}) = q'' \cdot R''_{h_o}$ can be written

and solved for $T_{s, O}$. This gives that $T_{s, O} = 1236 \text{ K}$. This means that the external temperature of the blade is above the allowable maximum temperature.

(b) Using figure 2 the thermal resistance network below can be drawn.



The only new thermal resistance is $R''_{k_{TBC}}$ and can be calculated by the equation

$$R''_{k_{TBC}} = \frac{L_{TBC}}{k_{TBC}} = 5 * 10^{-4} \frac{m^2 K}{W}.$$

Again R''_{tot} can be calculated by adding together all the thermal resistances and is found to be

$0.0047 \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$. Using the fact that $(T_{\infty,0} - T_{\infty,i}) = q'' * R''_{tot}$ q'' can be calculated to be

$320 \text{ kW} \cdot \text{m}^{-2}$. Using this the equation $(T_{\infty,0} - T_{IN,2}) = q'' * (R''_{h_o} + R''_{k_{TBC}} + R'')$ can be written

and solved for $T_{IN,2}$. This gives that $T_{IN,2} = 1104 \text{ K}$. This is under $T_{\max}$ and allows the turbine to be run safely.